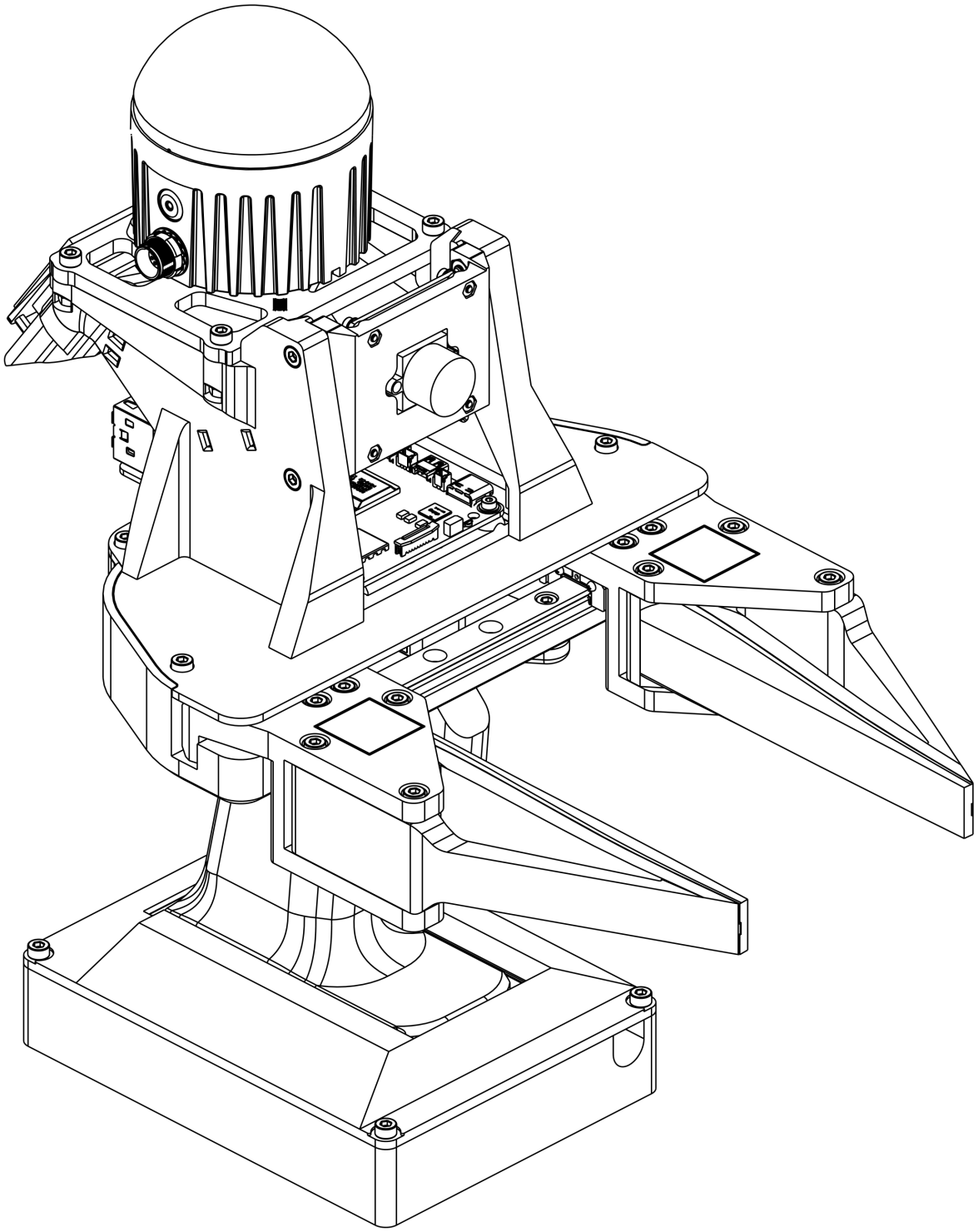
UMI-Live

Hardware Instructions



ASSEMBLY GUIDE

MECHANICAL / ELECTRONICS / WIRING

v8

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Notes

This guide assembles one UMI-Live gripper. Build two units for a bimanual setup.

Recommended Tools: M2/M3 drivers, scissors, flush cutters, multimeter, wire stripper.

3D Prints: remove and clean off all support materials.

Carriages: select two carriages that slide the smoothest on the rail.

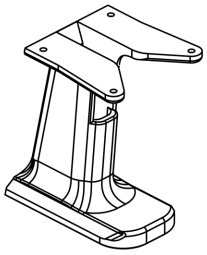
> **!** Keep carriages on a rail at all times to prevent the ball bearings from falling out.

Threadlocker: recommended for long-term or heavy use on metal-to-metal threads. Keep it off plastic, bearings, and sliding surfaces.

> **!** Keep the battery disconnected while assembling or changing wiring.

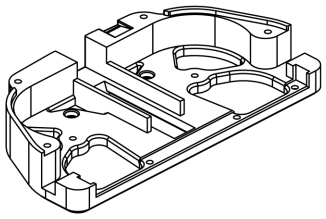
3D printed parts

grip



A **1x**

body_bottom



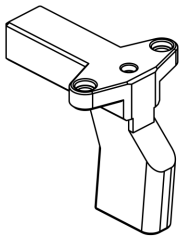
B **1x**

gear_axle



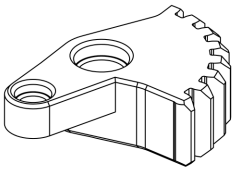
C **2x**

trigger



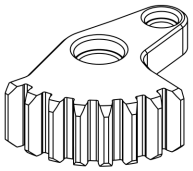
D **1x**

gear_left



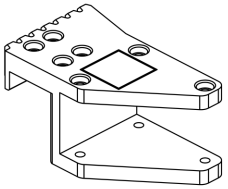
F **1x**

gear_right



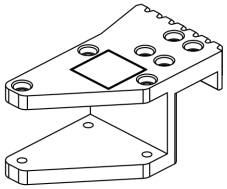
G **1x**

finger_holder_left



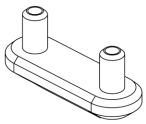
H-L **1x**

finger_holder_right



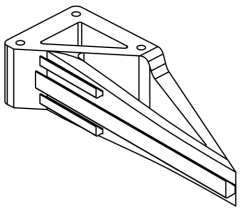
H-R **1x**

linkage



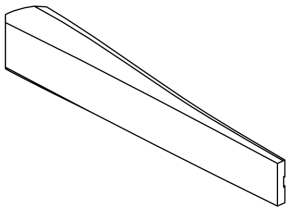
E **2x**

finger_backbone



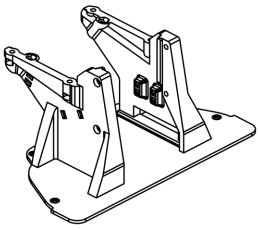
I **2x**

finger_pad



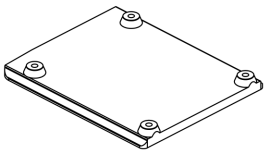
J **2x**

body_top



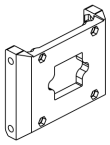
K **1x**

raspberry_slider



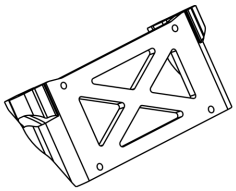
L **1x**

camera_mount



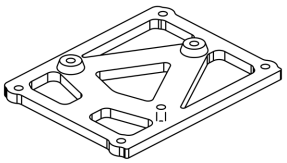
M **1x**

lcd_mount



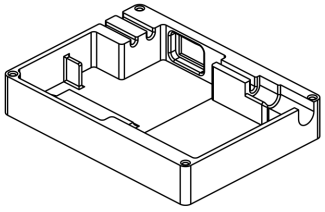
N **1x**

lidar_mount



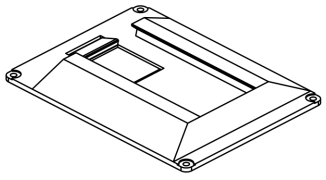
O **1x**

battery_pack_bottom



P **1x**

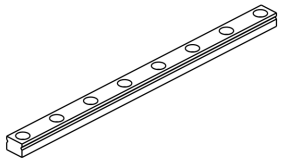
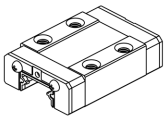
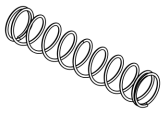
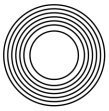
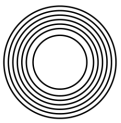
battery_pack_top



Q **1x**

Illustrations are not to scale.

Mechanical parts

MGN9 150mm Linear Rail  R **1x**	**MGN9C Carriage Blocks**  S **2x**	**uxcell MR85-2RS Ball Bearings**  T **8x**	**uxcell MR128-2RS Ball Bearings**  U **4x**
uxcell Compression Spring, 10x0.7x50mm  V **1x**	**ArUco tags (DICT_4X4_50, 16 mm)**  AM **2x**	**3M Gripping Material TB641, Black**  AN **Cut to fit**	**UHMW Polyethylene Transparent Film Slick Tape**  AP **Cut to fit**

Fasteners

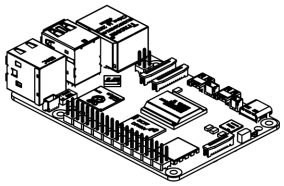
	H1 **M3 × 8 mm**	**14x**
	H2 **M3 × 12 mm**	**15x**
	H3 **M3 × 16 mm**	**2x**
	H4 **M3 × 20 mm**	**2x**
	H5 **M3 × 25 mm**	**10x**
	H6 **M3 × 30 mm**	**1x**
	H7 **M3 × 35 mm**	**6x**

	H8 **M3 nut**	**34x**
	H9 **M3 washer**	**2x**
	H10 **M2 × 6 mm**	**4x**
	H11 **M2 × 8 mm**	**8x**
	H12 **M2 nut**	**8x**
	H13 **M2 washer**	**12x**

Illustrations are not to scale.

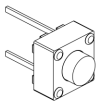
Electronics and cables

Raspberry Pi 5



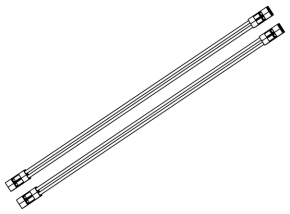
W **1x**

Tactile Button 2 Pin



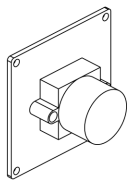
X **2x**

10cm Breadboard Jumper Wires



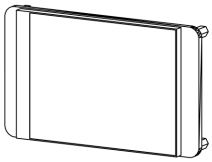
AL **4x**

SVPRO 2MP Global Shutter USB Camera



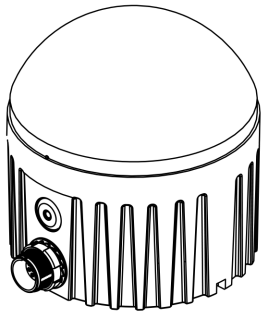
Y **1x**

Waveshare 2inch LCD Display SPI / 240 x 320



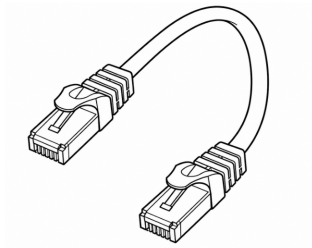
Z **1x**

RoboSense Airy LiDAR



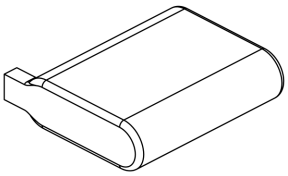
AA **1x**

Short Ethernet Cable 0.5FT



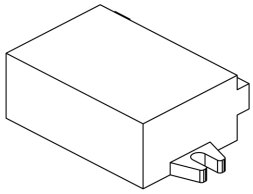
AK **1x**

12V 2600mAh Rechargeable Li-ion Battery Pack



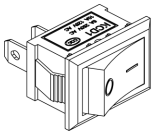
AB **1x**

DC-DC 12V/24V to 5V 5A 25W Buck Converter USB-C output



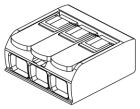
AC **1x**

DaierTek Mini ON and Off Rocker Switch



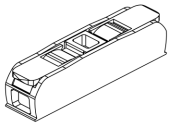
AD **1x**

WAGO 221-413 Lever Nuts



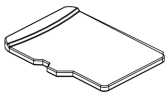
AF **2x**

WAGO 221-2401 Lever Nuts



AG **1x**

microSD card

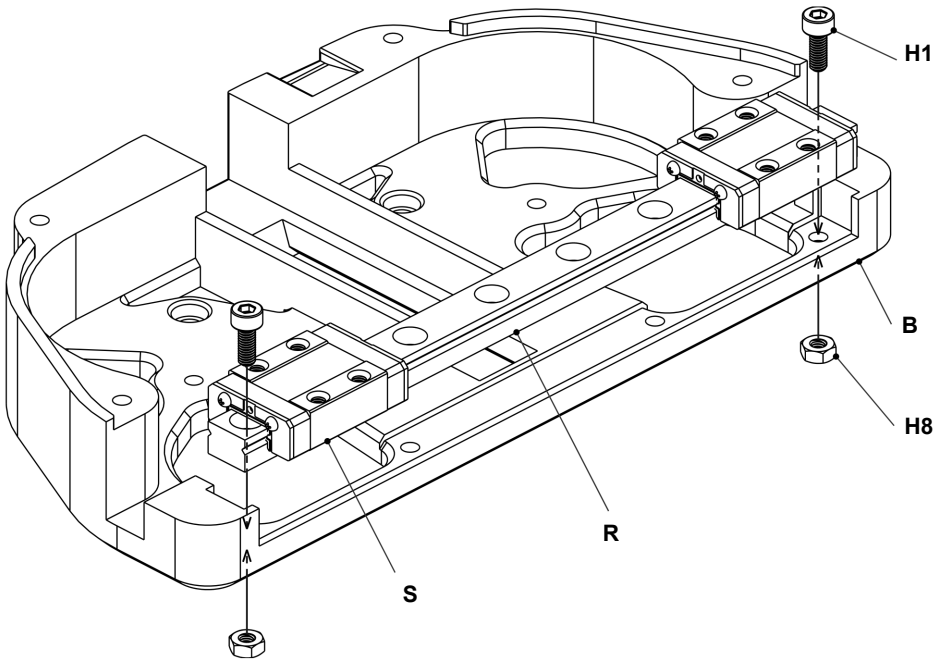


AH **1x**

Illustrations are not to scale.

STEP 1

Fit the rail and carriages



Select the two smoothest carriage blocks. Secure the rail to the body bottom using two M3 x 8 screws in the outer holes and two nuts underneath.

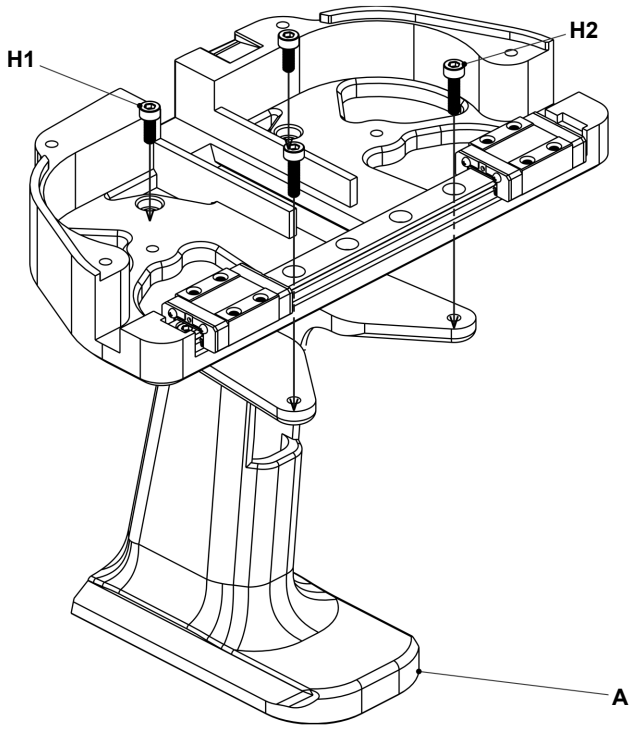
Keep both carriages on the rail to prevent ball bearings from falling out.

PARTS ADDED

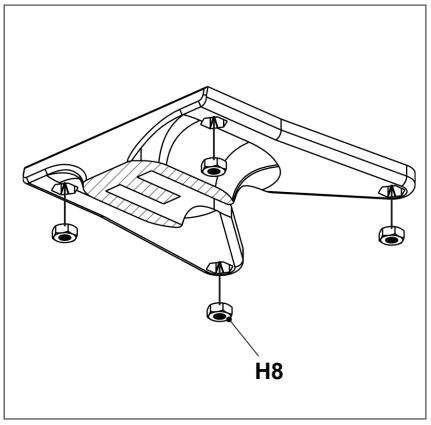
B	**1x**
Body bottom	
R	**1x**
Linear rail	
S	**2x**
Carriage block	
H1	**2x**
M3 × 8 mm	
H8	**2x**
M3 nut	

STEP 2

Attach the grip



UNDERSIDE VIEW



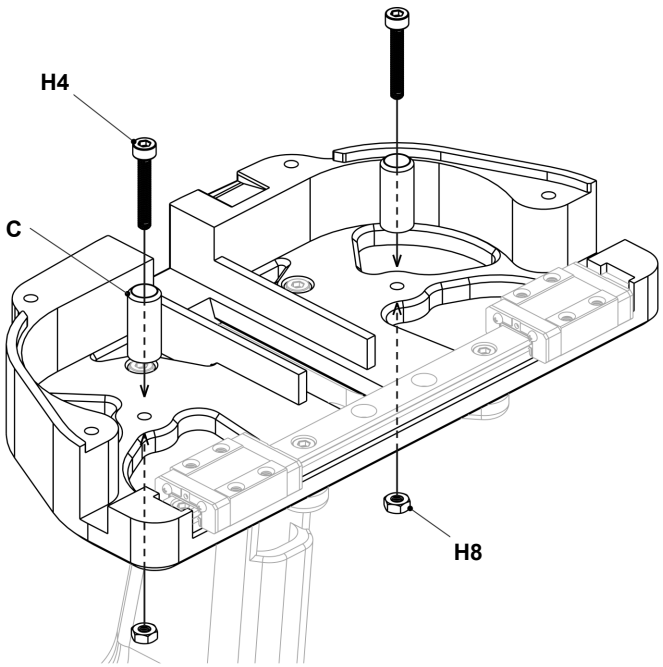
Align the grip beneath the body bottom. Fit two M3 x 12 screws through the inner rail holes, body bottom, and grip, plus two M3 x 8 screws through the other grip mounting holes. Secure all four screws with nuts underneath and tighten evenly.

PARTS ADDED

A	**1x**
Grip	
H1	**2x**
M3 × 8 mm	
H2	**2x**
M3 × 12 mm	
H8	**4x**
M3 nut	

STEP 3

Install the two gear axles



Secure each gear axle with one M3 x 20 screw and nut. Ensure that both axles are seated vertically.

PARTS ADDED

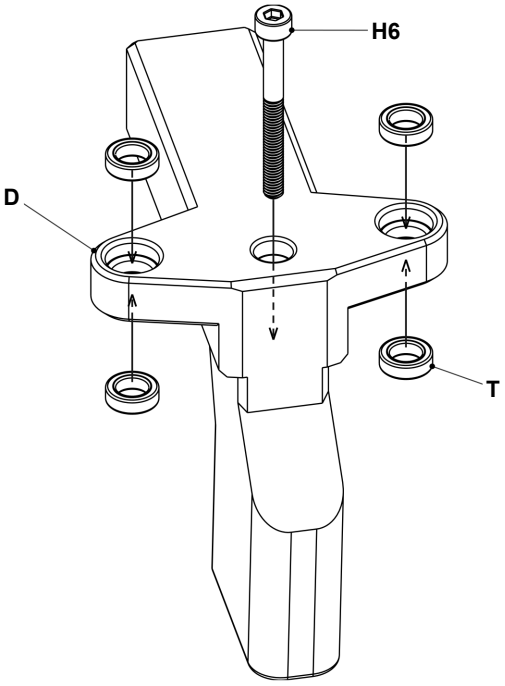
C **2x**
Gear axle

H4 **2x**
M3 × 20 mm

H8 **2x**
M3 nut

STEP 4

Prepare the trigger



Seat one small bearing on each face of both trigger pivots (four total). Screw the M3 x 30 screw into the central hole, threading the plastic.

Do not force the bearings into their recesses. If the fit is too tight, carefully file to adjust the trigger's bearing holes until the bearings seat without excessive force.

PARTS ADDED

D **1x**
Trigger

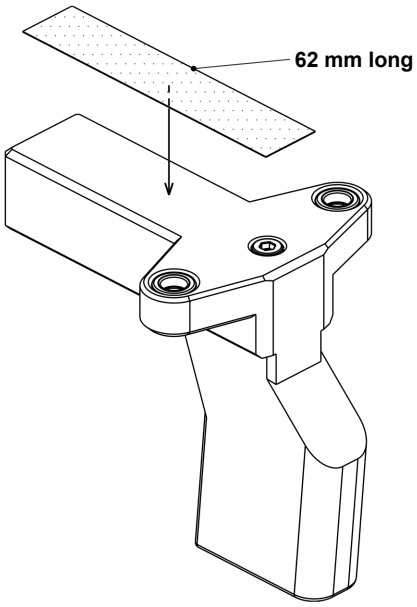
T **4x**
Small bearing

H6 **1x**
M3 × 30 mm

STEP 5

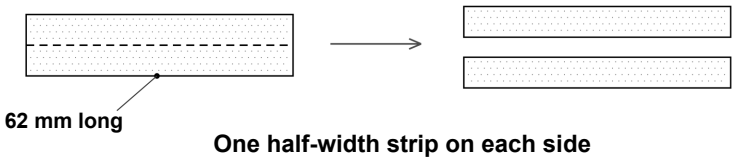
Apply low-friction tape

1. Top strip

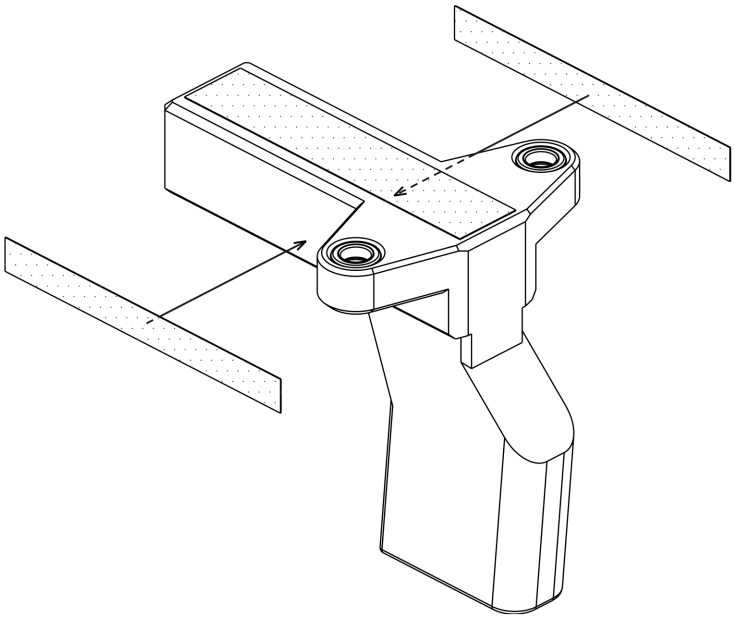


Cut a 62 mm long strip of low-friction tape. Apply it along the top of the trigger, covering the recessed screw.

2. Side strips



One half-width strip on each side



Cut another 62 mm long strip of low-friction tape, and cut it lengthwise. Apply one half-width strip to each side of the trigger.

Press firmly over all applied tape to secure it.

Do not stack or overlap tape layers. Apply tape only to flat surfaces; do not bend around corners.

PARTS ADDED

AP **Cut to fit**

Low-friction tape

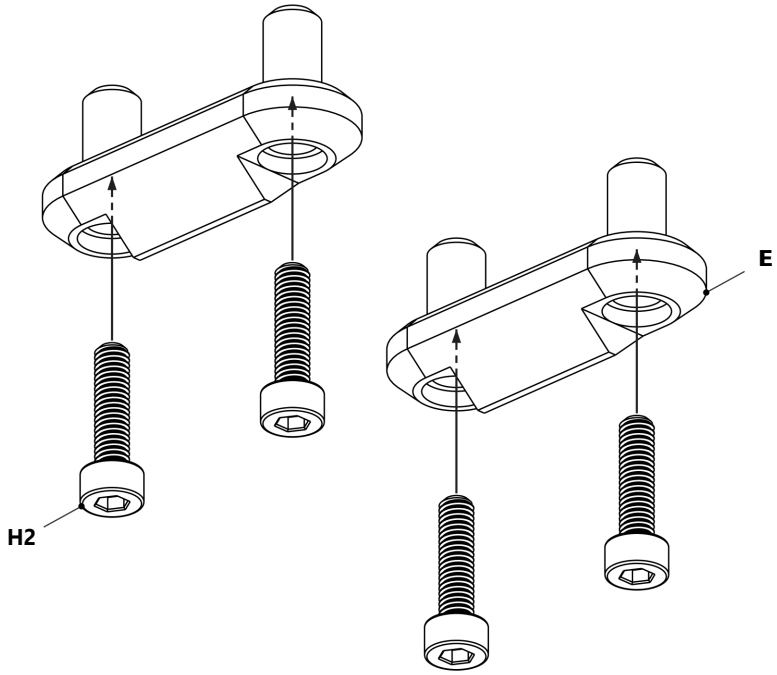
ALREADY ASSEMBLED

Prepared trigger

From Step 4

STEP 6

Prepare both linkages



Screw two M3 x 12 screws into each linkage from its underside, threading the plastic.

PARTS ADDED

E **2x**

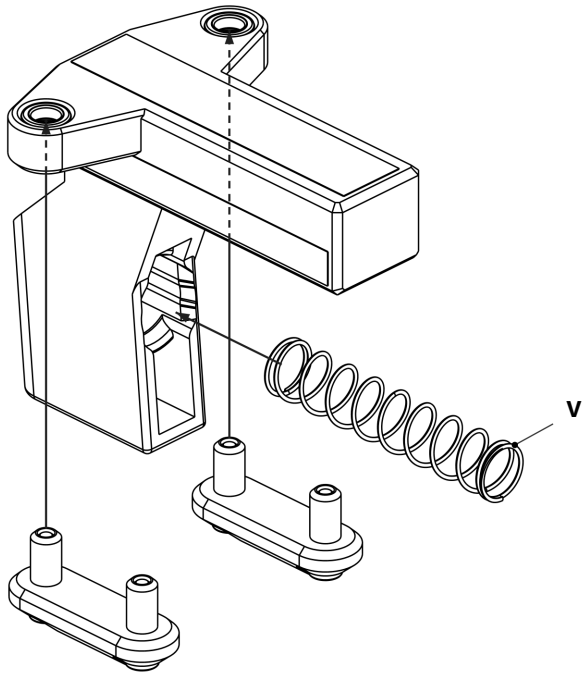
Linkage

H2 **4x**

M3 × 12 mm

STEP 7

Join the trigger and linkages



Bring both prepared linkages up from below into the trigger bearings. Slide the compression spring into the cylindrical slot in the trigger, along the axis shown.

PARTS ADDED

V **1x**

Compression spring

ALREADY ASSEMBLED

Prepared trigger

From Step 5

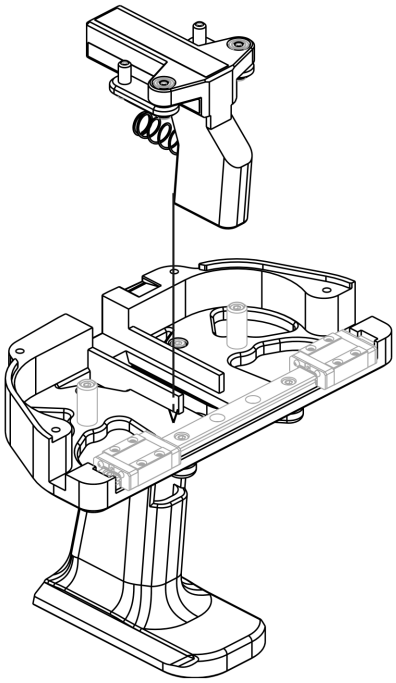
Prepared linkages

From Step 6

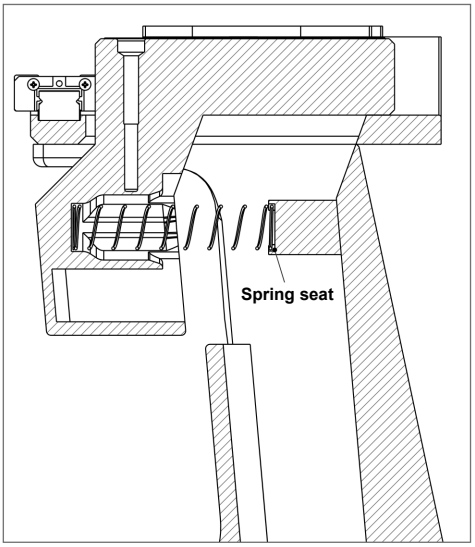
STEP 8

Seat the trigger in the body

LOWER INTO BODY



CROSS-SECTIONAL VIEW



Lower the trigger into its slot in the body assembly. Position the spring into its seat in the grip. Ensure both linkages remain fully seated in their bearings.

Tweezers may be needed to position the spring correctly in its seat.

ALREADY ASSEMBLED

Body assembly

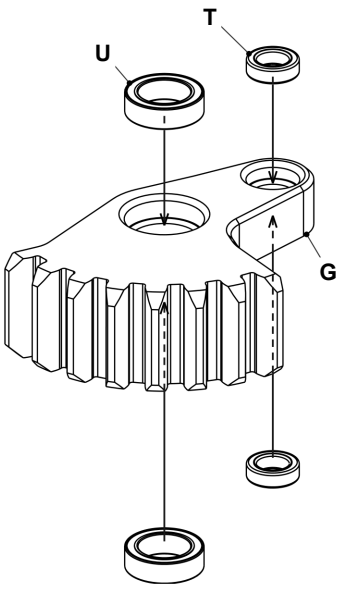
From Step 3

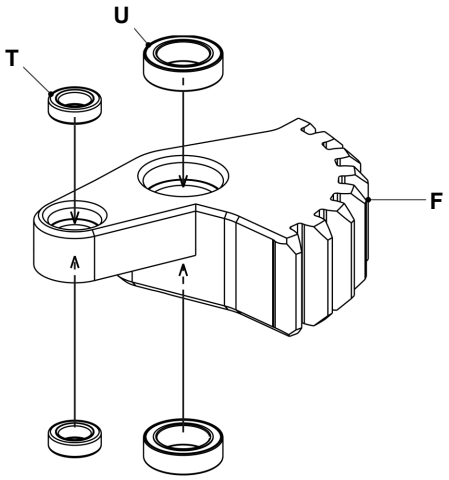
Trigger + linkages

From Step 7

STEP 9

Prepare the left and right gears





Fit two large MR128-2RS bearings into the main pivot and two small MR85-2RS bearings into the linkage pivot of each gear. Ensure that bearings are seated squarely in their recesses.

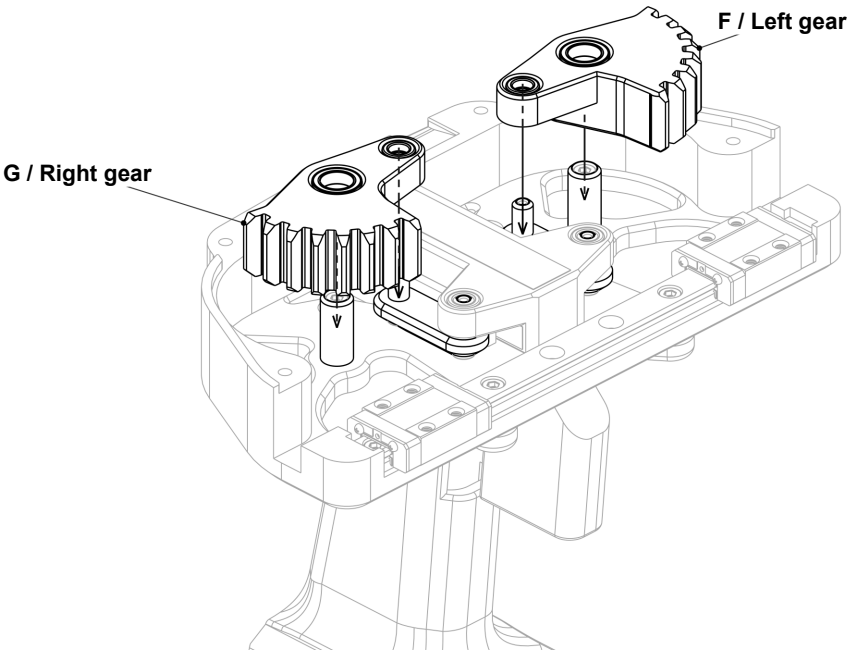
Do not force the bearings into their recesses. If the fit is too tight, carefully file to adjust the gear holes until the bearings seat without excessive force.

PARTS ADDED

F	**1x**
Left gear	
G	**1x**
Right gear	
U	**4x**
Large bearing	
T	**4x**
Small bearing	

STEP 10

Install the gears



Place the prepared gears over the two printed axles. Ensure that all bearings are fitted into their axles. Move the trigger gently to check that both sides travel together smoothly.

ALREADY ASSEMBLED

Body assembly

From Step 8

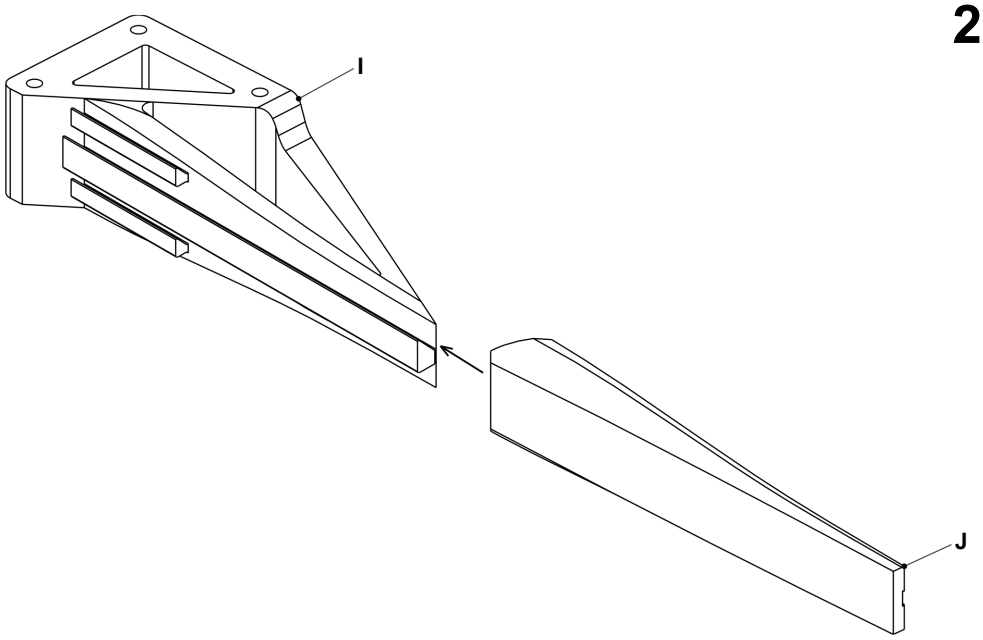
Prepared gears

From Step 9

STEP 11

Fit the TPU finger pads

2x



Slide the TPU pad onto its backbone from the finger tip. Keep it aligned with the backbone as you push/pull it along the full length. Build another set for a total of two fingers.

The pad slides on from the front; do not push it on sideways.

PARTS ADDED

I **2x**

Finger backbone

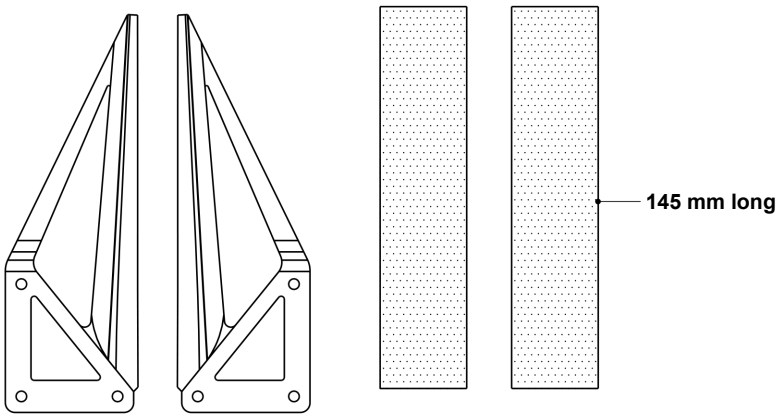
J **2x**

Finger pad

STEP 12

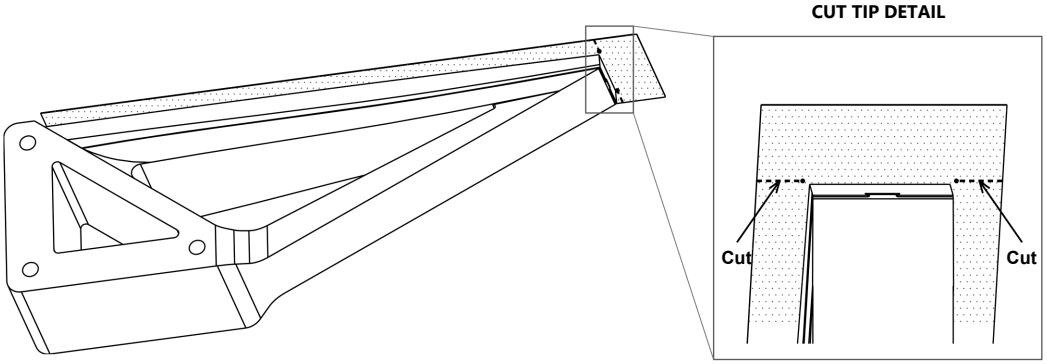
Apply the grip tape

1. Ready for tape



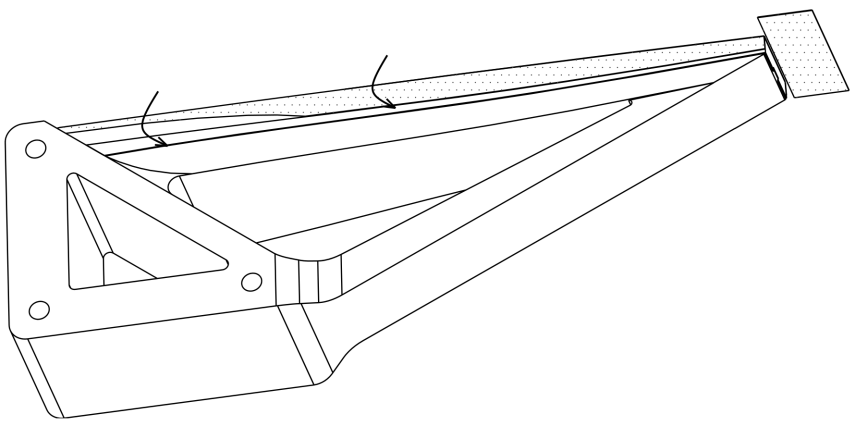
Start with both padded fingers from Step 11. Cut two strips of grip tape, each around 145 mm long.

2. Apply the tape and cut the tip



Center the tape on the TPU contact face, leaving overhangs along both long edges and the tip. Cut inward from each tape edge to the tip corner (detail), leaving the center uncut.

3. Fold the long edges



Fold both long edges around the TPU pad and press them down. Keep the tip tab free for the next substeps.

PARTS ADDED

AN **Cut to fit**

Grip tape

ALREADY ASSEMBLED

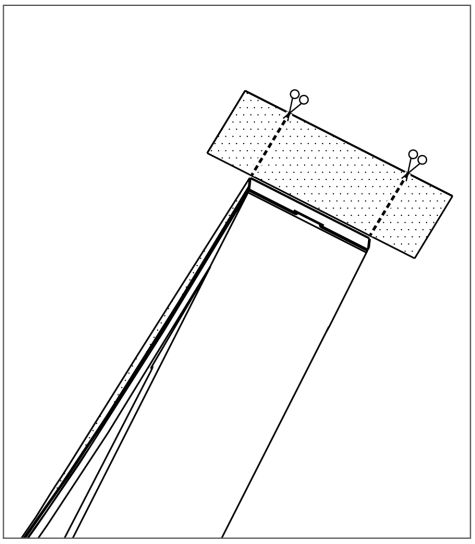
Finger pair

From Step 11

STEP 12 CONT.

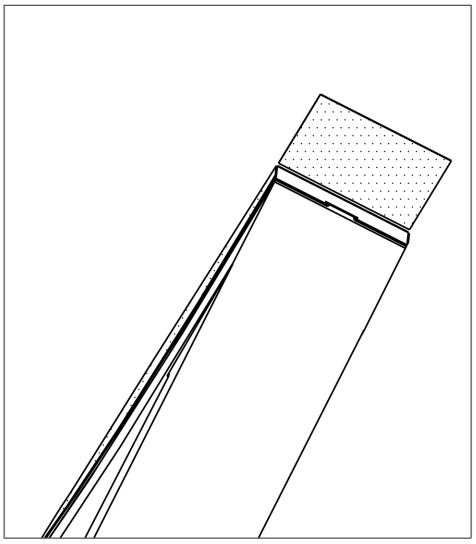
Finish the grip tape

4. Trim the tip sides



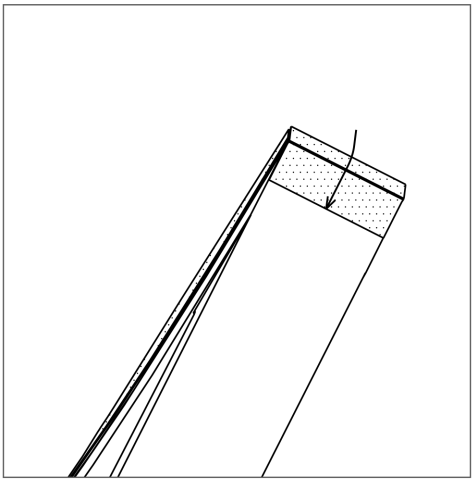
Trim the tab sides along the dashed lines so that each cut is aligned with an edge of the finger.

5. Tab trimmed to width



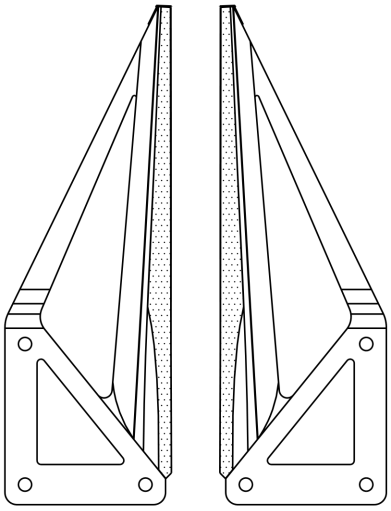
Check that the tab now matches the width of the finger. Ensure this flap is long enough to fold over the tip.

6. Fold over the tip



Fold the flap over the tip and press it flat against the back of the finger.

7. Finished pair



Repeat for the other finger.

Press firmly over all applied tape to secure it.

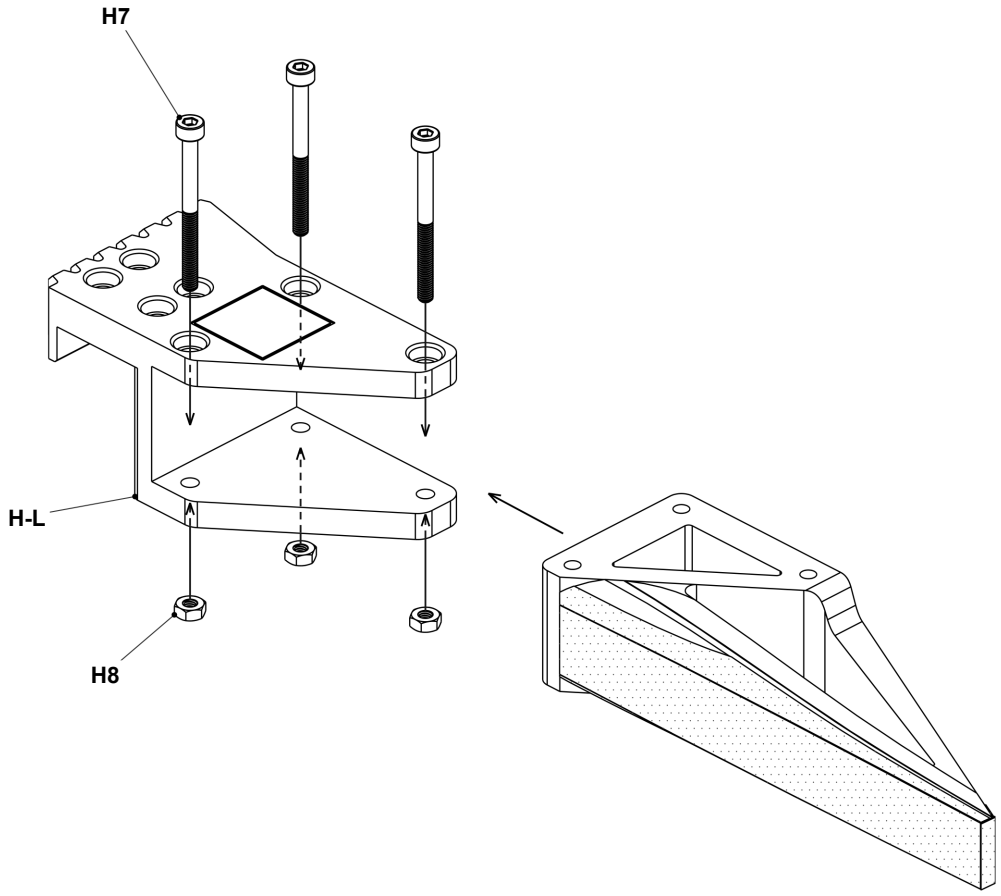
CONTINUE WITH

Same fingers and grip tape from the previous page.

STEP 13

Attach fingers to holders

2x



Slide one completed finger into the left holder (H-L), as shown. Align the three holes and fit three M3 × 35 mm fasteners and three nuts. Repeat with the right holder (H-R) and the other finger.

Fit the ArUco tags

Print the supplied tag sheet at 100% (Actual size), with Fit and Shrink disabled. Check the 50 mm scale before cutting.

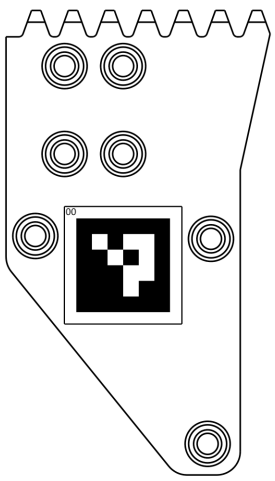
Dictionary: DICT_4X4_50

Black marker: 16 × 16 mm

Cut size: 20 × 20 mm; keep the 2 mm white margin.

Attach one tag inside each holder's recessed border using double-sided tape. Ensure the tag is flat and secured.

Use two distinct IDs; assign left and right in software.



PARTS ADDED

H-L **1x**
Left finger holder

H-R **1x**
Right finger holder

H7 **6x**
M3 × 35 mm

H8 **6x**
M3 nut

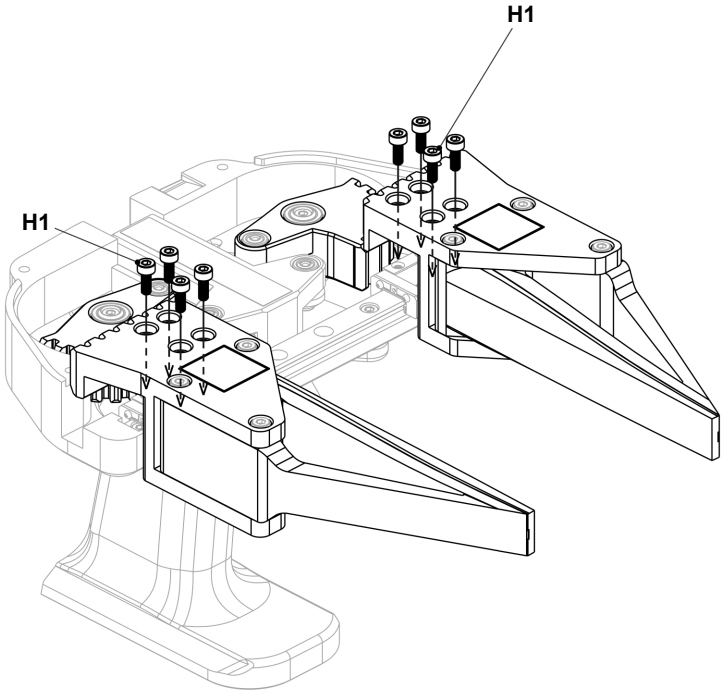
AM **2x**
ArUco tags

ALREADY ASSEMBLED

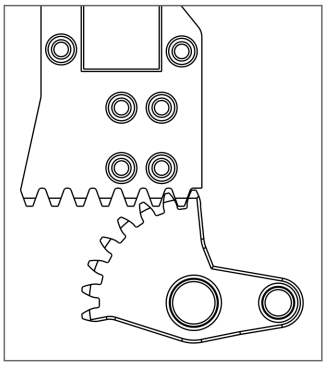
Finger pair
From Step 12

STEP 14

Attach both finger assemblies



TOP VIEW OF LEFT SIDE



Mesh the left holder (H-L) with the left gear (F), and the right holder (H-R) with the right gear (G). Align each holder over its carriage and secure it with four M3 × 8 mm fasteners. Check that both fingers move symmetrically.

PARTS ADDED

H1 **8x**

M3 × 8 mm

ALREADY ASSEMBLED

Body assembly

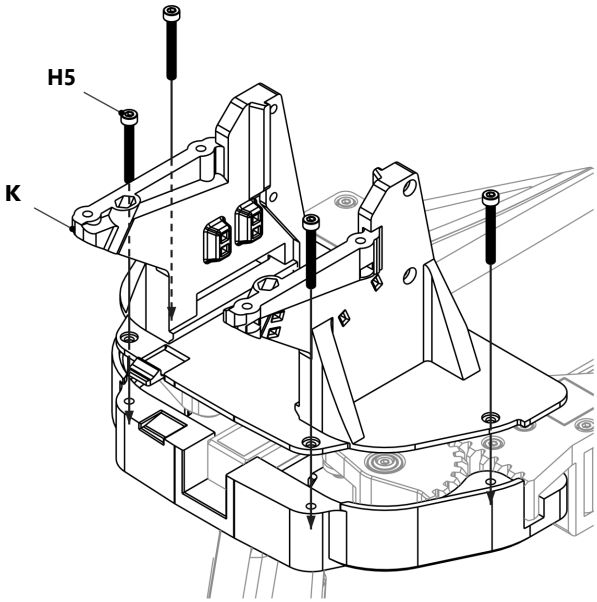
From Step 10

Finger assemblies

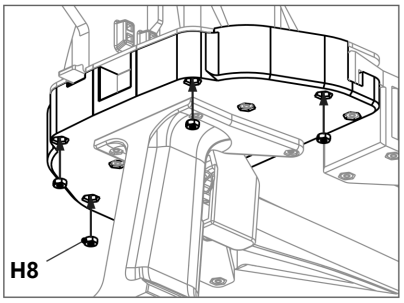
From Step 13

STEP 15

Close the body



UNDERSIDE VIEW



Lower the body top over the completed mechanism. Fit four M3 x 25 screws and nuts. Tighten gradually, checking the trigger and finger movement to prevent binding.

PARTS ADDED

K **1x**

Body top

H5 **4x**

M3 × 25 mm

H8 **4x**

M3 nut

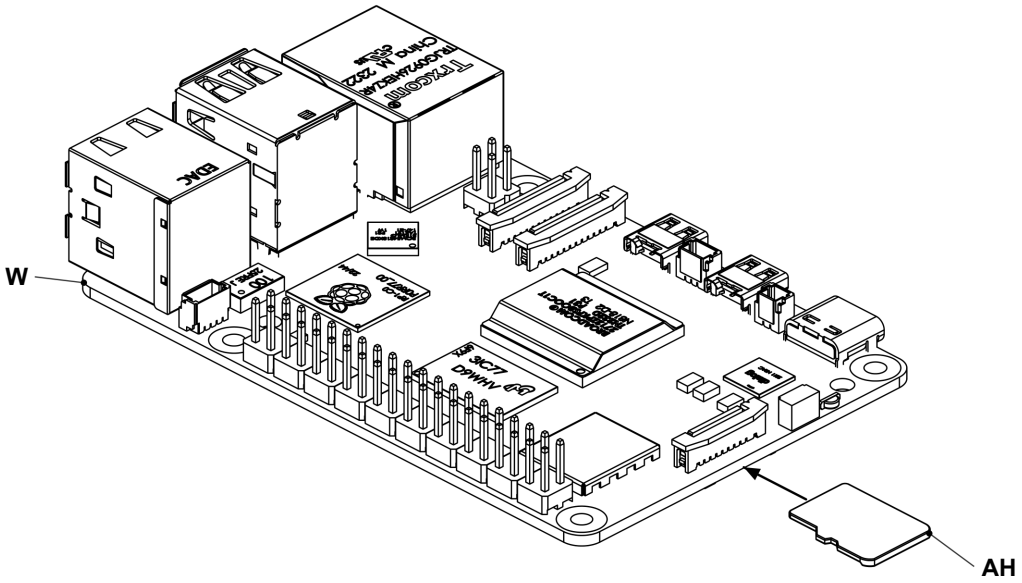
ALREADY ASSEMBLED

Body assembly

From Step 14

STEP 16

Prepare the Pi



Prepare the microSD card using the UMI-Live software setup guide before first use. Software setup can be completed after assembly.

Insert the microSD card into the underside slot, with the gold contacts facing up toward the board.

Slide it in gently; do not force it.

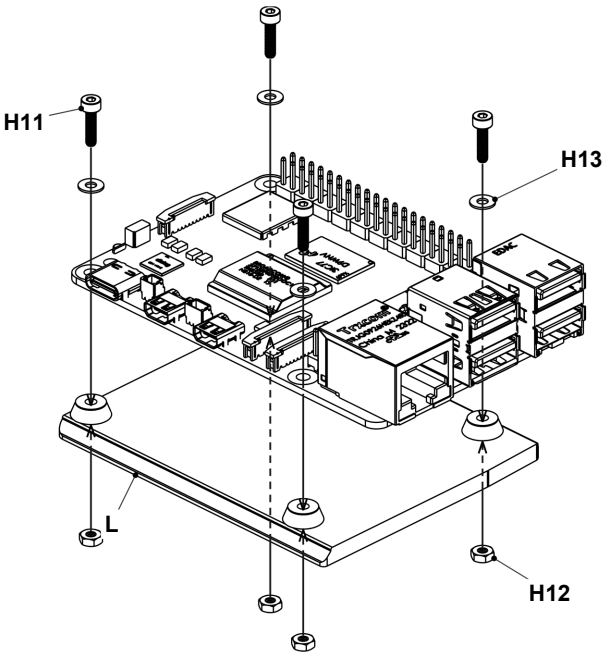
PARTS ADDED

W **1x**
Raspberry Pi 5

AH **1x**
microSD card

STEP 17

Mount the Pi on its slider



Place the Raspberry Pi on its slider and align the four mounting holes. Fit four M2 x 8 screws, washers, and nuts. Tighten just enough to secure the Raspberry Pi. Do not overtighten.

Note the orientation of the slider and Pi.

PARTS ADDED

L **1x**
Pi slider

H11 **4x**
M2 × 8 mm

H12 **4x**
M2 nut

H13 **4x**
M2 washer

ALREADY ASSEMBLED

Prepared Pi
From Step 16

STEP 18

Connect the wires to the Pi

PARTS ADDED

AL **4x**

Dupont jumper wires

Female-to-female **1x**

LCD wire

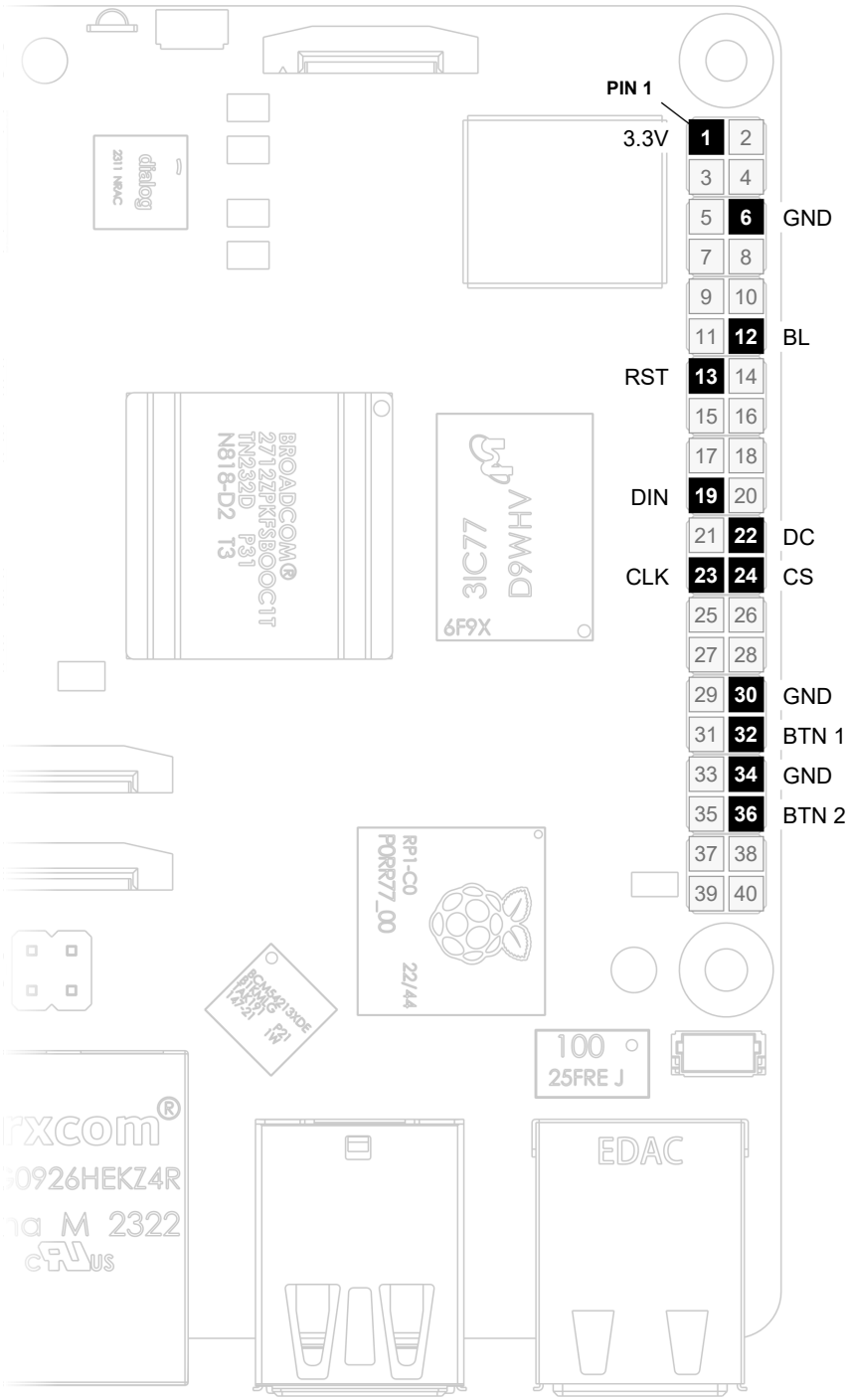
Included with LCD

1. Plug the supplied cable into the LCD. Trace each wire from its printed connector label to the loose end.
2. Connect all eight LCD leads to the Pi using the table and pin diagram, then unplug the LCD end.
3. Connect four female-to-female Dupont leads to the Pi for BTN 1 and BTN 2. Keep each pair together.

GPIO numbers are different from physical pin numbers.

Keep power disconnected.

PIN HEADER



CONNECTIONS

LCD	PI SIGNAL	PIN
VCC	3.3V	1
GND	Ground	6
DIN	GPIO10	19
CLK	GPIO11	23
CS	GPIO8	24
DC	GPIO25	22
RST	GPIO27	13
BL	GPIO18	12

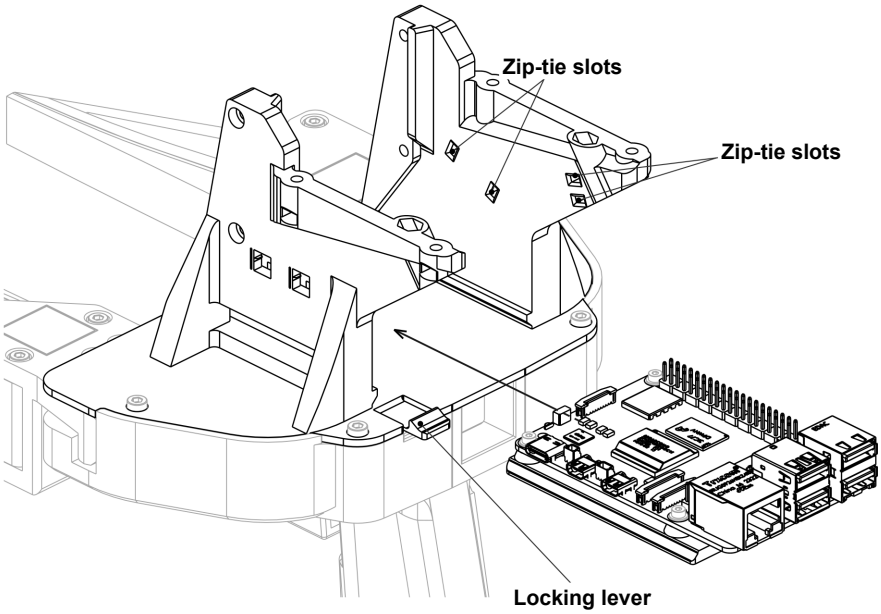
BTN 1	PI SIGNAL	PIN
Signal	GPIO12	32
GND	Ground	30

BTN 2	PI SIGNAL	PIN
Signal	GPIO16	36
GND	Ground	34

Wires are not shown in the CAD illustrations.

STEP 19

Slide the Pi into the body



Insert the Pi and slider forward into the body top guides until the lever clicks and locks. Thread optional zip ties through the marked slots for LCD cable management; leave them loose until wiring is complete.

PARTS ADDED

2 optional

Zip ties

ALREADY ASSEMBLED

Body assembly

From Step 15

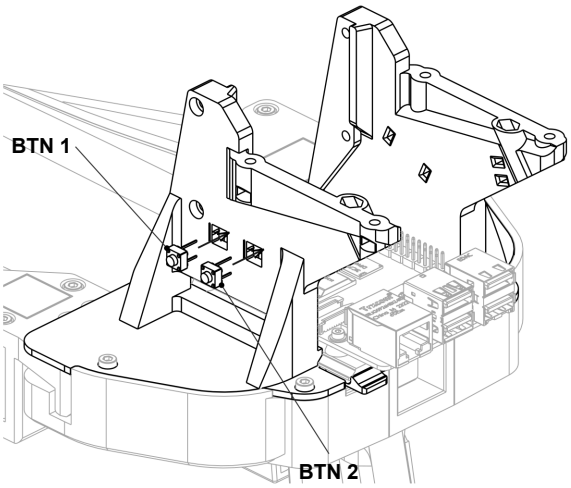
Pi + slider

From Step 17

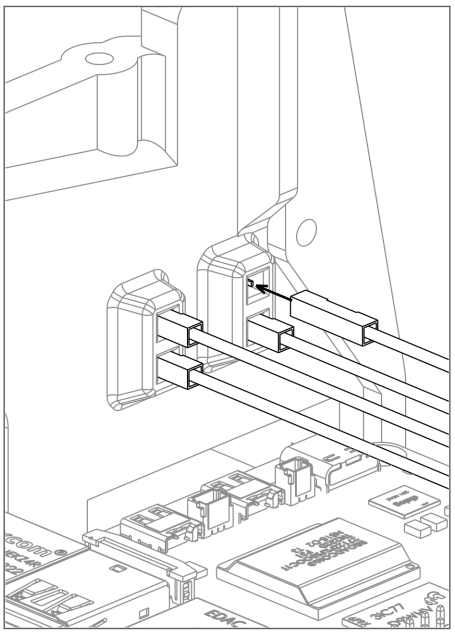
STEP 20

Install the buttons

FIT THE BUTTONS



CONNECT THE WIRES



Fully press the buttons in the side openings; the tabs are meant to bend to tightly fit the buttons. Connect the button nearest the fingers to GPIO12 and GND (physical pins 32 and 30). Connect the button nearest the LCD to GPIO16 and GND (physical pins 36 and 34). Refer to the diagram in Step 18 as needed.

PARTS ADDED

X **2x**

Tactile button

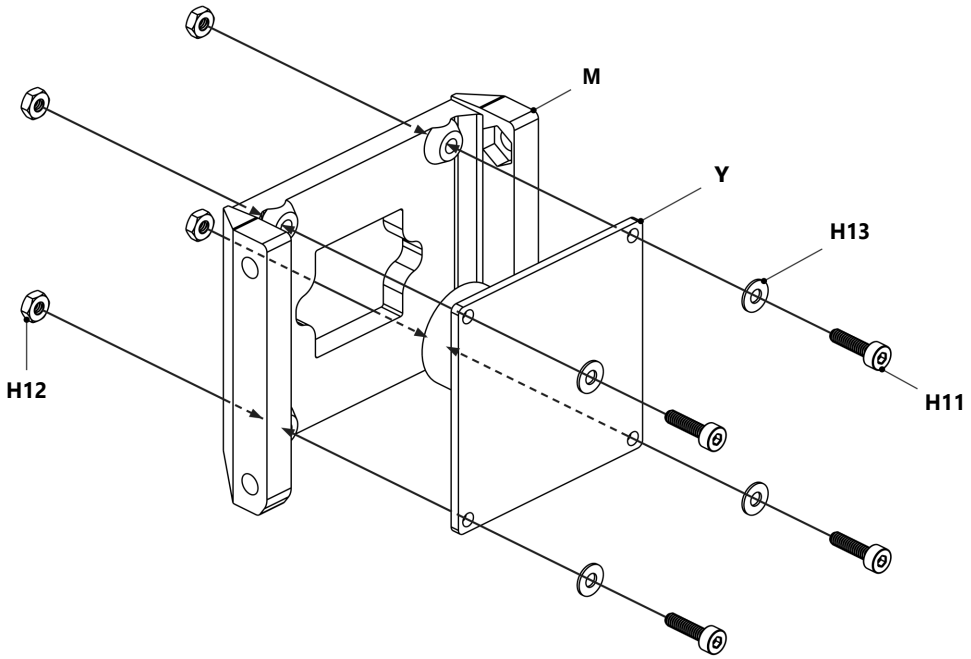
ALREADY ASSEMBLED

Body assembly

From Step 19

STEP 21

Mount the camera board



Bring the camera board forward from behind the mount, with the lens facing through the opening. Fit four M2 x 8 screws, washers and nuts. Connect the 4-pin JST PH end of the USB camera cable to the camera board.

PARTS ADDED

M **1x**
Camera mount

Y **1x**
SVPRO USB camera

From camera **1x**
USB Camera Cable

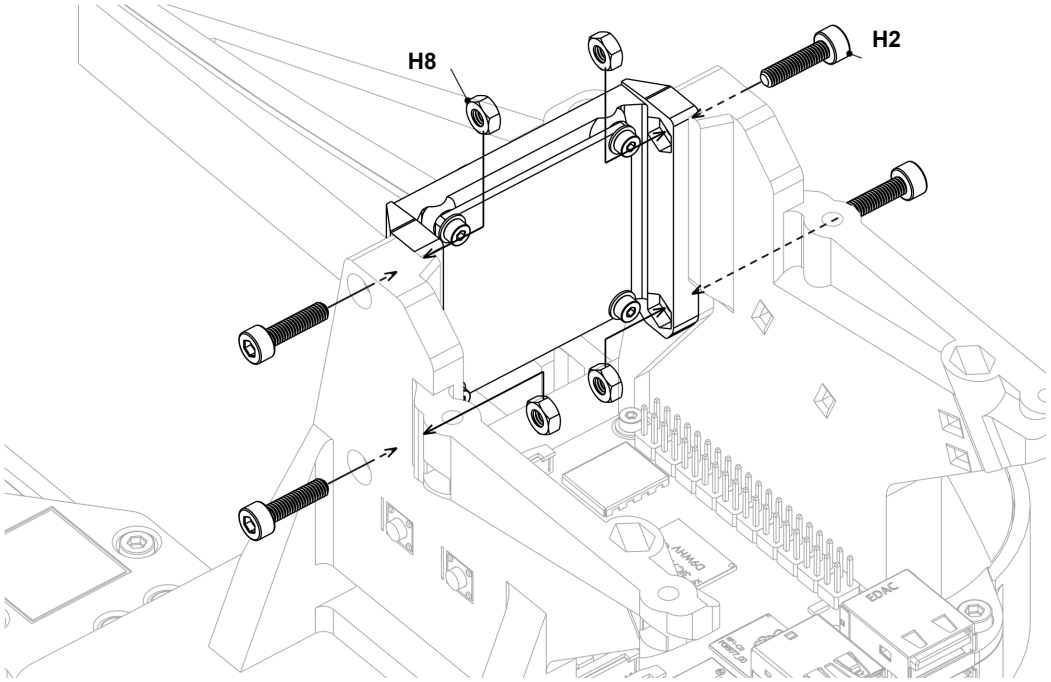
H11 **4x**
M2 × 8 mm

H12 **4x**
M2 nut

H13 **4x**
M2 washer

STEP 22

Attach and connect the camera



Secure the camera mount to the front of the body top with four M3 x 12 screws and nuts. Plug the USB camera cable into a blue USB-A 3.0 port on the Raspberry Pi.

PARTS ADDED

H2 **4x**
M3 × 12 mm

H8 **4x**
M3 nut

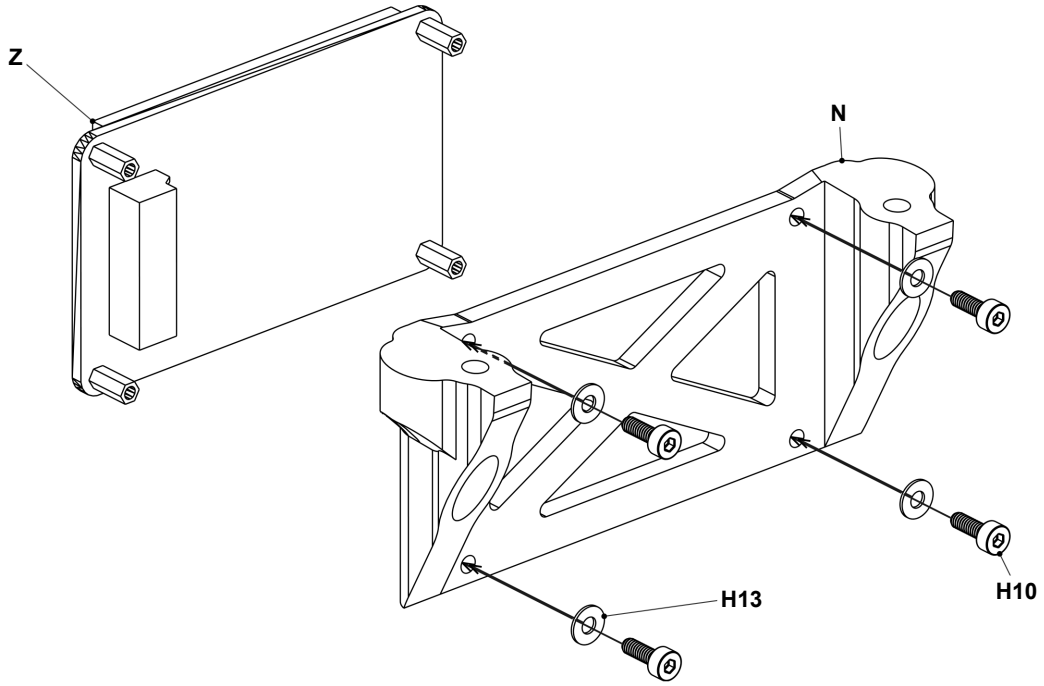
ALREADY ASSEMBLED

Body assembly
From Step 20

Camera assembly
From Step 21

STEP 23

Fit the LCD to its mount



Align the LCD onto the printed mount with the display facing outward. Fit four M2 x 6 screws and washers.

PARTS ADDED

N **1x**
LCD mount

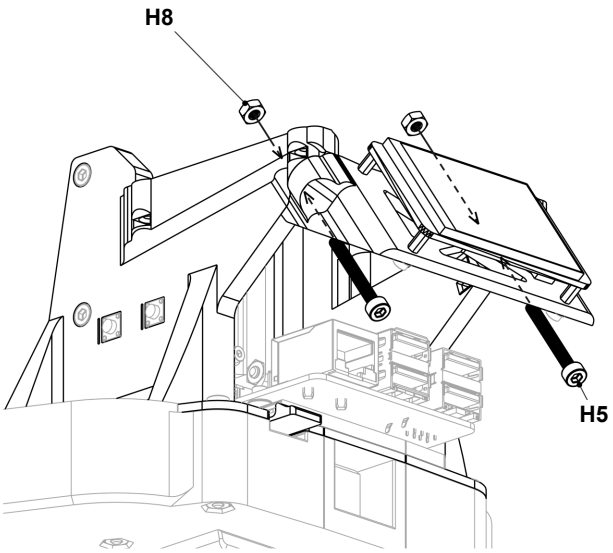
Z **1x**
Waveshare LCD

H10 **4x**
M2 × 6 mm

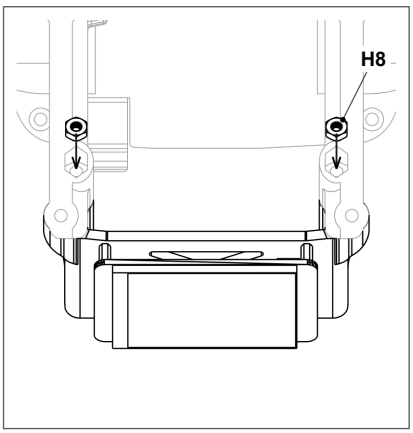
H13 **4x**
M2 washer

STEP 24

Attach and connect the LCD



TOP VIEW



Mount the LCD assembly at the rear of the body using two M3 x 25 screws and nuts. Connect the LCD cable from the Pi to the LCD display. It is recommended to recheck each cable before applying power.

Ensure that the LCD cable is fully pushed into the LCD display.

PARTS ADDED

H5 **2x**
M3 × 25 mm

H8 **2x**
M3 nut

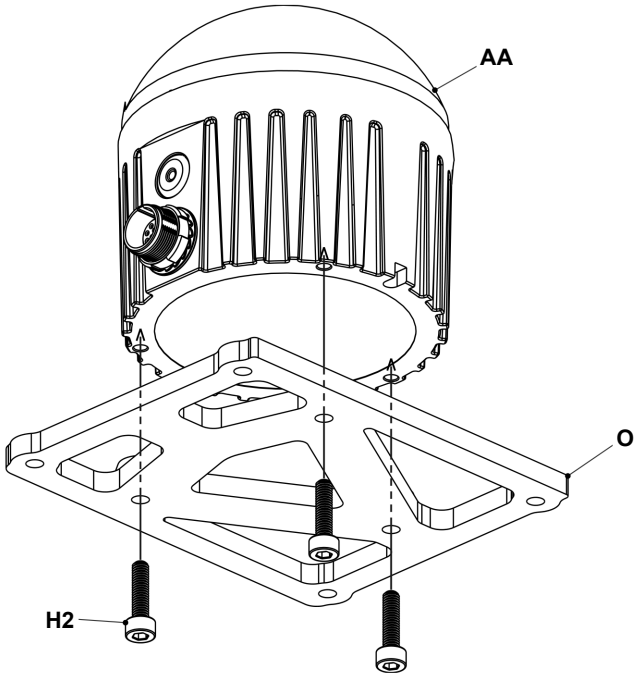
ALREADY ASSEMBLED

Body assembly
From Step 22

LCD assembly
From Step 23

STEP 25

Fasten Airy to the LiDAR mount



Align the Airy LiDAR base with the LiDAR mount. Note the orientation of the LiDAR cable connector. Screw three M3 x 12 screws through the plate into the LiDAR.

Avoid touching the LiDAR optical dome. Keep the dome and ventilation clear.

PARTS ADDED

O **1x**

LiDAR mount

AA **1x**

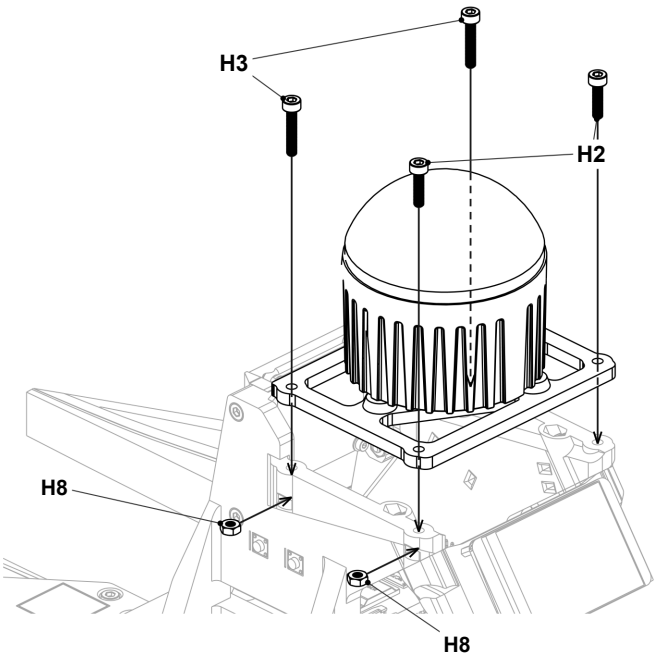
RoboSense Airy LiDAR

H2 **3x**

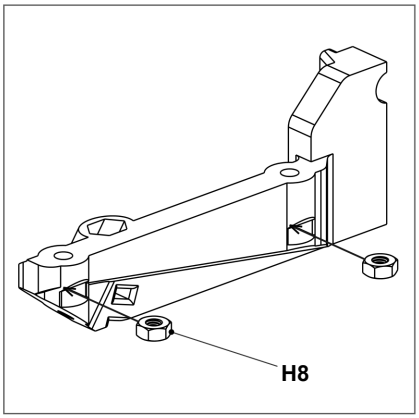
M3 × 12 mm

STEP 26

Install the LiDAR assembly



OPPOSITE SIDE



Slide two M3 nuts inward from each outer side into the four slots. Place the LiDAR mount on the supports. Fit M3 x 16 screws at the front holes (camera side) and M3 x 12 screws at the rear (LCD side).

PARTS ADDED

H2 **2x**

M3 × 12 mm

H3 **2x**

M3 × 16 mm

H8 **4x**

M3 nut

ALREADY ASSEMBLED

Body assembly

From Step 24

LiDAR assembly

From Step 25

STEP 27 Prepare wires

Shortening is optional and makes the battery pack more organized. If shortening, cut to the lengths below.

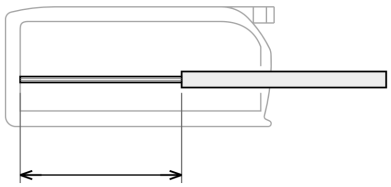
Strip 11 mm from every WAGO end, then twist the bare strands. Optionally solder the tip to prevent fraying.

> **!** Do not short circuit the battery! Keep the insulator tubes on the bare wire leads when not in use.

COMPONENT	LEAD	CUT LENGTH
Battery	Red	29 mm
	Black	85 mm
Switch	Lead 1	28 mm
	Lead 2	49 mm
Converter	Red	65 mm
	Black	63 mm

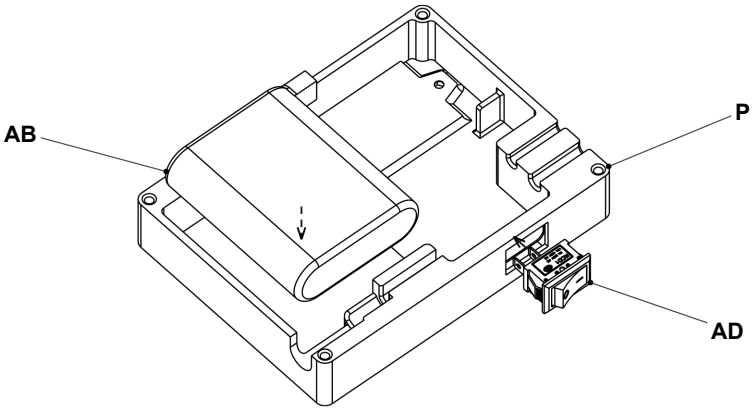
Total cut length includes the 11 mm stripped end.

WAGO SIDE VIEW



11 mm stripped

STEP 28 Fit the battery and switch



Keep the battery disconnected and the switch OFF. Seat the battery in the large compartment. Push the switch into the side opening until it clips into place.

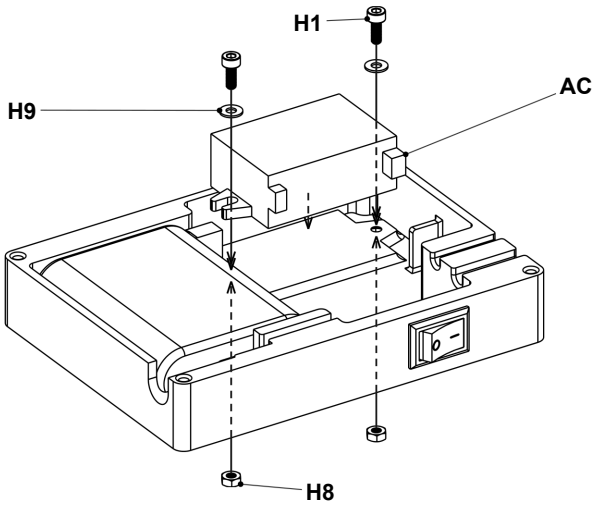
PARTS ADDED

P **1x**
Battery pack bottom

AB **1x**
Battery

AD **1x**
Power switch

STEP 29 Secure the converter



Place the converter beside the battery. Secure it with two M3 x 8 mm screws, placing a washer under each screw head. The two nuts go beneath the case.

PARTS ADDED

AC **1x**
DC-DC converter

H1 **2x**
M3 x 8 mm screw

H8 **2x**
M3 nut

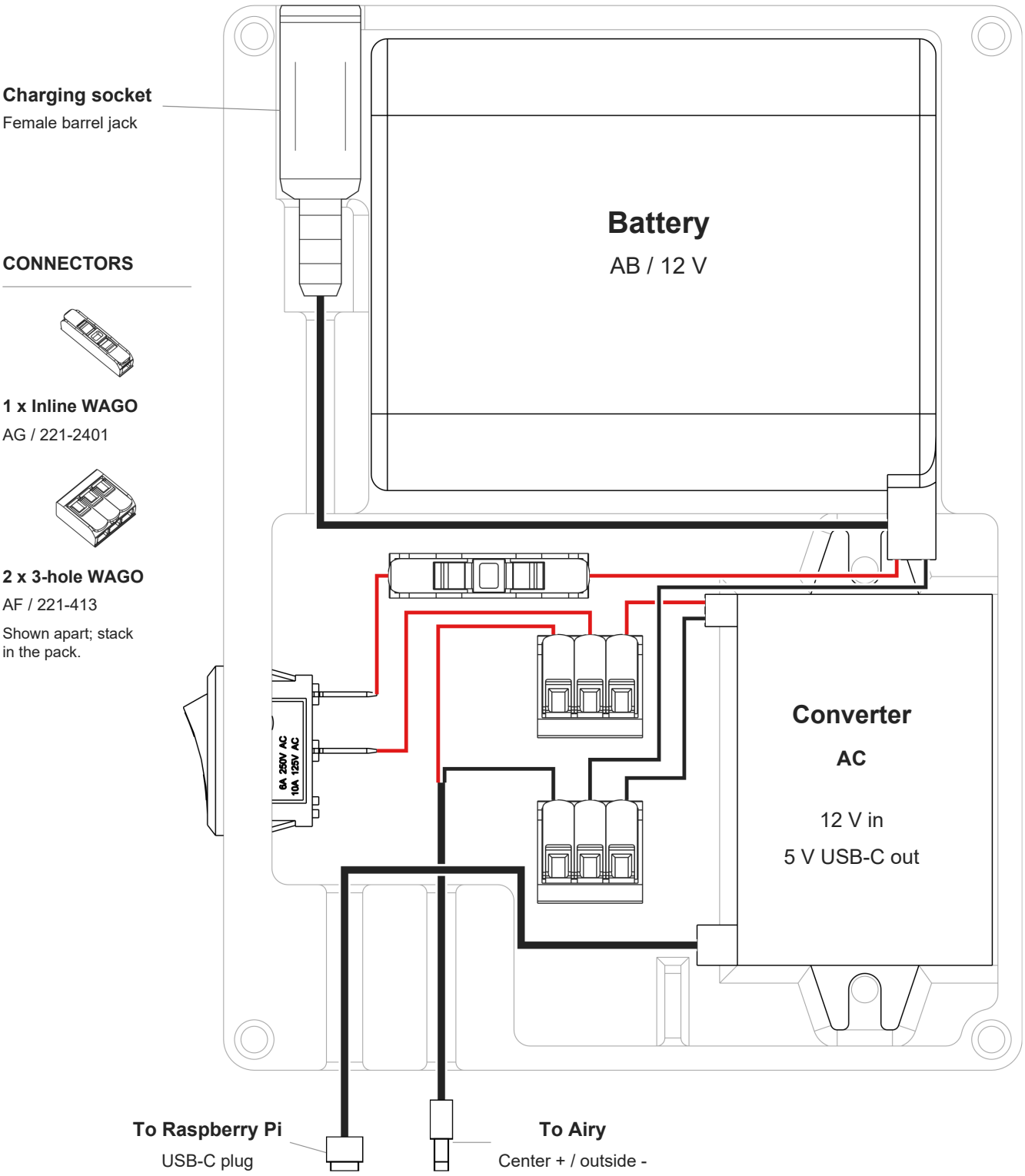
H9 **2x**
M3 washer

ALREADY ASSEMBLED

Open battery pack
From Step 28

STEP 30

Wire the battery pack



Positive (+) Negative (-) Gaps at wire overlaps = no connection.

Connect one wire per hole: strip to the marked length and gather all strands. Optional anti-fraying: solder only the very tip; leave the clamped section unsoldered. Lift the lever, insert fully, close, then gently tug-check. All three holes of each 221-413 connect internally.

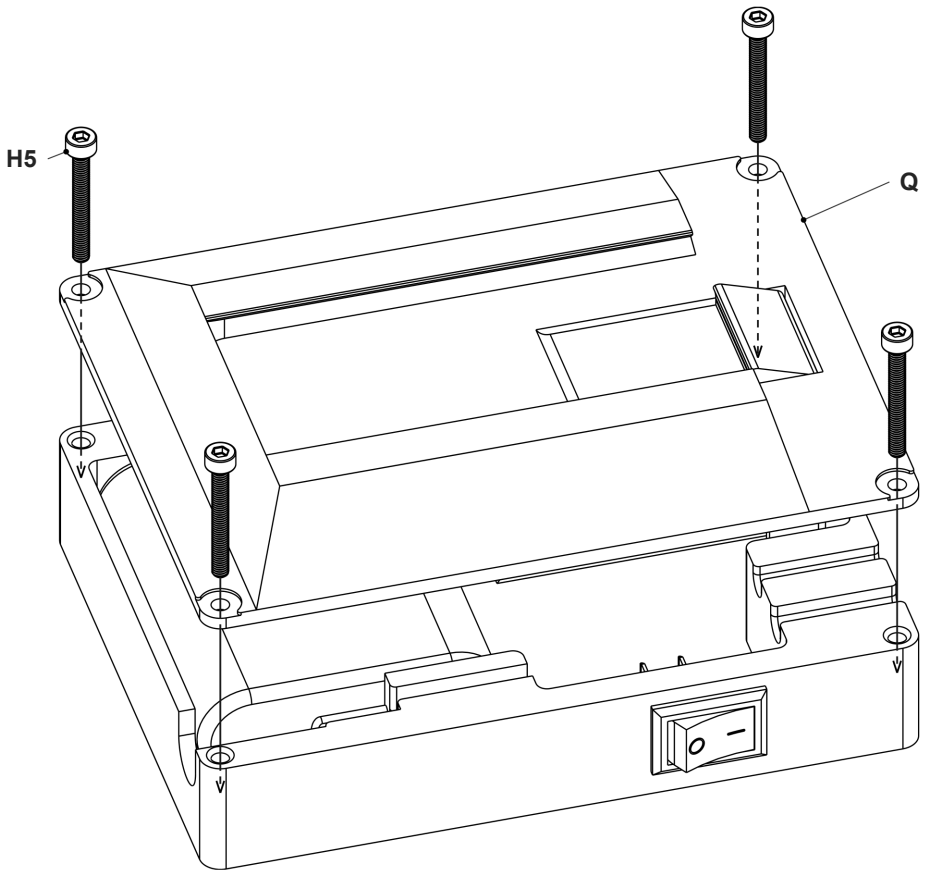
Output test: leave the Pi and Airy unplugged. Connect the battery and switch ON. Set the multimeter to DC volts: red probe in the barrel center, black probe on the outside. Expect about +12 V. Then switch OFF and disconnect the battery.

STEP 31

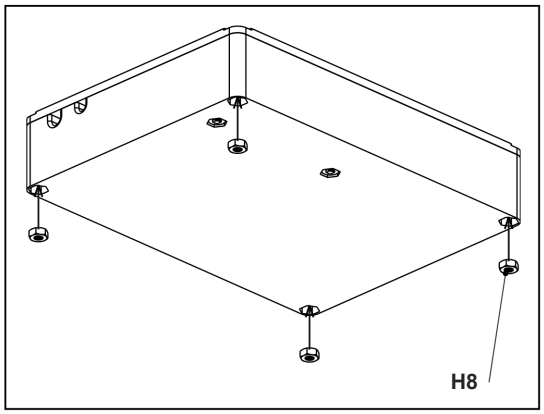
Close the battery pack

Finish the wiring and output test on page 23 before closing the case.

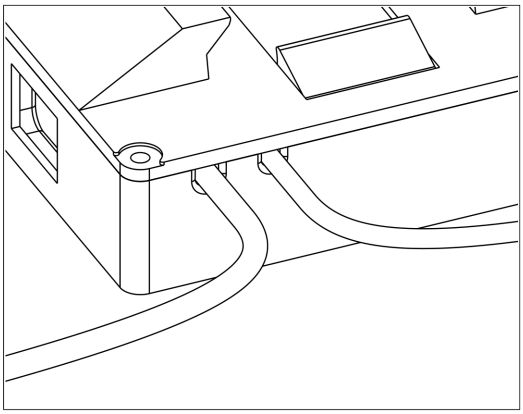
FIT THE COVER



NUT POCKETS / UNDERSIDE



CABLE EXITS / LID SEATED



1. After the output test passes, keep the switch OFF and reconnect the battery. Put each output cable into its case outlet.

2. Lower the lid carefully. Keep every wire away from the lid edge, screw posts and screw holes.

3. Fit four M3 x 25 mm screws and four nuts. Check that the switch is accessible and both cables are held securely.

PARTS ADDED

Q **1x**

Battery pack top

H5 **4x**

M3 x 25 mm screw

H8 **4x**

M3 nut

ALREADY ASSEMBLED

Wired battery pack

From Step 30

STEP 32

Slide on the battery pack

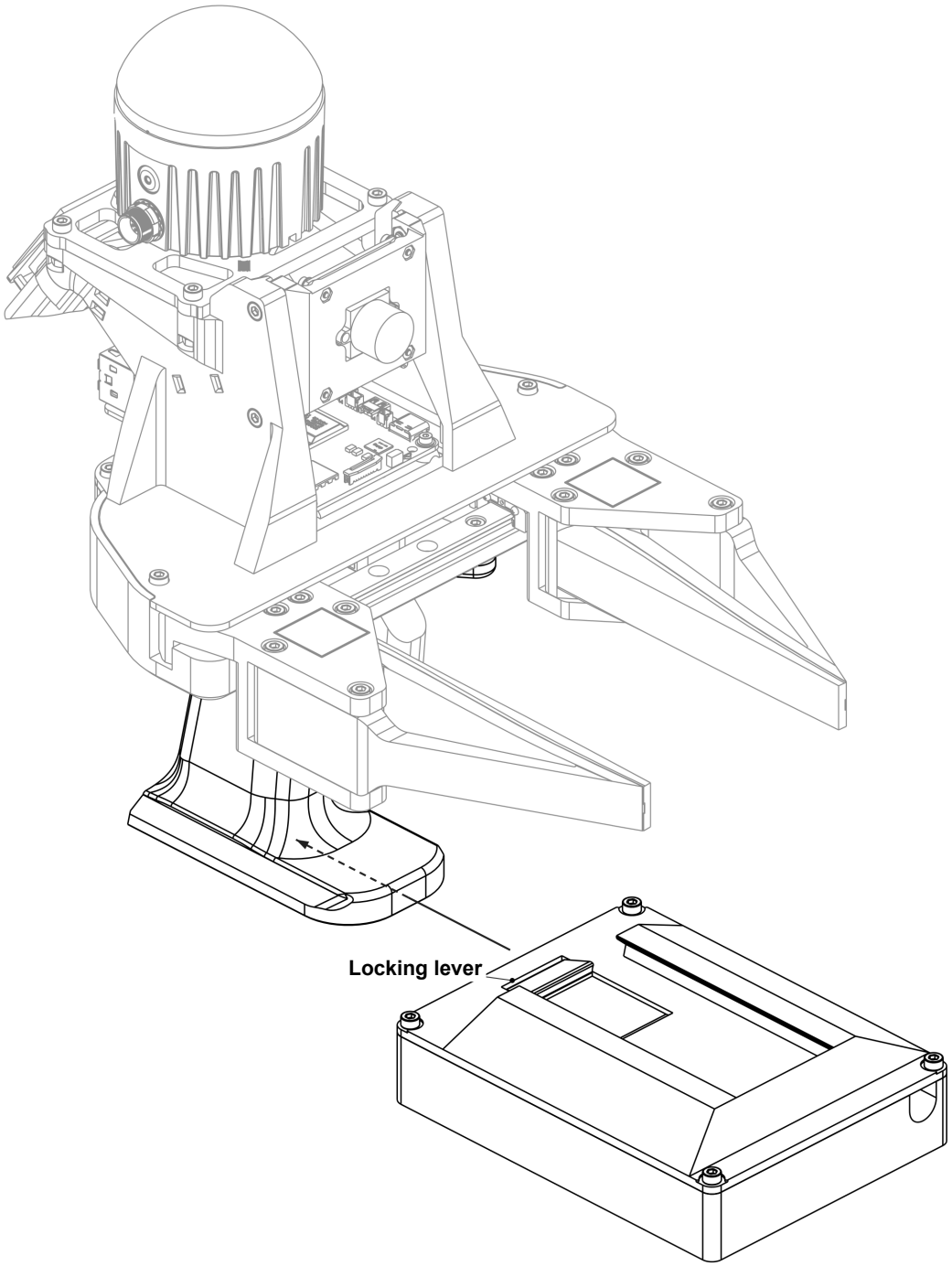
ALREADY ASSEMBLED

UMI-Live assembly

From Step 26

Battery pack

From Step 31

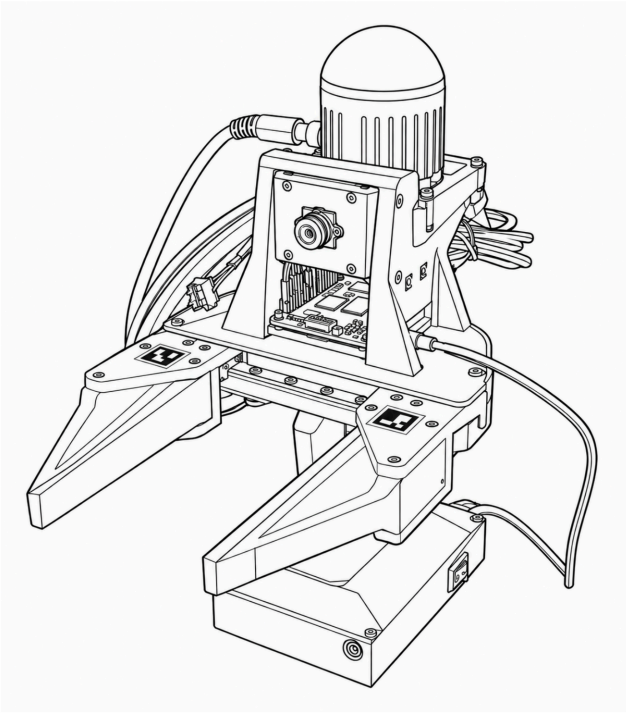


Align the battery-pack rails with the base of the grip.
Slide the pack until it clicks and locks into place.

If the battery pack lever is unable to be pushed down, open the battery pack and ensure that the components inside are not obstructing its motion.

STEP 33

Connect all wires



CABLE ROUTING REFERENCE

Illustrated from the UMI-Live repository photo.

Keep the switch OFF while connecting the cables.

1. Hand-tighten the interface cable onto the LiDAR connector.
2. Connect the Ethernet cable between the interface cable and the Pi Ethernet port.
3. Connect the pack DC barrel output to the interface cable power input.
4. Connect the pack USB-C output to the Pi power input.
5. Check the camera USB-A 3.0 connection, LCD cable and button leads.

PARTS ADDED

AK **1x**

Ethernet cable

From LiDAR **1x**

Interface cable

ALREADY ASSEMBLED

Completed hardware

From Step 32

Final check

- ☐ Trigger returns under spring force; both fingers move smoothly together.
- ☐ Fasteners are seated; bearings and carriages remain free.
- ☐ Tags are flat, visible and have distinct IDs.
- ☐ Buttons use pins 32/30 and 36/34. LCD follows the pin map.
- ☐ Camera connects to USB 3.0; Airy connects to Ethernet and its original power cable.
- ☐ No bare conductors, trapped wires or cable loops in the moving mechanism.
- ☐ Pack output test on page 23 passed; both assemblies are fully seated.

Charge the battery

1. Switch OFF. Disconnect all connections to the battery, including both bare-wire leads from the WAGO harness.
2. Plug the supplied charger directly into the battery's charging barrel socket.
3. Unplug the charger before reconnecting the battery to the harness.

Do not charge and use the battery at the same time.